# MATH PROBLEMS.PNG - Solution

## Number Theory ### 1. Find all integers $x, y$ with $x^2 + 2 = y^3$. To solve the equation $x^2 + 2 = y^3$, we consider the possible values of $y$: - If $y = 0$, then $x^2 + 2 = 0^3 = 0$, which gives $x^2 = -2$, not possible for integer $x$. - If $y = 1$, then $x^2 + 2 = 1^3 = 1$, which gives $x^2 = -1$, not possible. - If $y = 2$, then $x^2 + 2 = 2^3 = 8$, which gives $x^2 = 6$, not an integer square. - If $y = 3$, then $x^2 + 2 = 3^3 = 27$, which gives $x^2 = 25$, so $x = \pm 5$. For $y = 3$, we have solutions $(x, y) = (5, 3)$ and $(-5, 3)$. - For $y = 4$, $x^2 + 2 = 4^3 = 64$, giving $x^2 = 62$, not an integer square. - For $y = 5$, $x^2 + 2 = 5^3 = 125$, giving $x^2 = 123$, not an integer square. Checking further values of $y$ quickly becomes infeasible, but we notice that $x^2 = y^3 - 2$ implies $x^2$ is close to a cube, and the differences grow larger as $y$ increases. Thus, the only solutions are $(x, y) = (5, 3)$ and $(-5, 3)$. ### 2. Prove there are infinitely many $n$ such that $n \mid (2^n - 2)$ and $n$ is composite. Consider $n = 2^k - 1$ for $k \geq 2$. By Fermat's Little Theorem, $2^k \equiv 2 \pmod{k}$, implying $k \mid (2^k - 2)$. If $k$ is composite, then $n = 2^k - 1$ is composite. For example, $k = 4$ gives $n = 15$, and $k = 6$ gives $n = 63$. Since there are infinitely many composite numbers $k$, there are infinitely many such $n$. ### 3. Determine all $n \in \mathbb{N}$ for which $\varphi(n) \mid n - 1$. The Euler's totient function $\varphi(n)$ counts the integers up to $n$ that are coprime to $n$. We need $\varphi(n) \mid n - 1$. - For $n = 2$, $\varphi(2) = 1 \mid 1$. - For $n = 3$, $\varphi(3) = 2 \mid 2$. - For $n = 4$, $\varphi(4) = 2 \nmid 3$. - For $n = 5$, $\varphi(5) = 4 \mid 4$. Checking small values suggests that $n$ being a prime or power of a prime might work. Indeed, if $n = p^k$ for a prime $p$, then $\varphi(n) = p^k - p^{k-1}$. For $k = 1$, $\varphi(p) = p - 1 \mid p - 1$. For $k > 1$, $\varphi(p^k) = p^k - p^{k-1} = p^{k-1}(p - 1)$, and $p^{k-1}(p - 1) \mid p^k - 1$ since $p^k - 1 = (p - 1)(p^{k-1} + \cdots + 1)$. Thus, $n = p^k$ works for any prime $p$ and $k \geq 1$. ### 4. Solve in integers: $x^4 + 4y^4 = z^2$. Rewrite the equation as $z^2 = x^4 + 4y^4 = (x^2 + 2xy + 2y^2)(x^2 - 2xy + 2y^2)$. For $z$ to be an integer, both factors must be perfect squares. Set $x^2 + 2xy + 2y^2 = a^2$ and $x^2 - 2xy +